

A ternary five-subset network and a conditional integer-multiplication saving of 2^{-28}

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Abstract

We replace the binary triple-based interchange motif by a five-subset motif over $\mathbb{F}_3$, retain its pair totals, and resynthesize common-four-point sum circuits. The resulting rational interface certifies interchange saving $3/400000000$. A paired weighted-triple producer also improves the complex interface to saving $39/10^9$. Combined with the existing compact-control and fast Gaussian resampling arguments, the exact assembly parameters give $\kappa = 373/10^{11} > 2^{-28}$ with a strict absorption gap. All unchanged interfaces of the pinned upstream manuscript remain assumptions.

1 Scope, prior work and attribution

The computational model is a fixed finite alphabet and a fixed number of one-dimensional tapes. The source is OpenAI’s *Integer multiplication below $n \log n$* , pinned at commit

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The original theorem, compact-control movement, reservation, repair, stopped guard, numerical error analysis and final rounding are retained assumptions. Finite certificates do not constitute a full formalization of that theorem.

This is Zhihao Chen’s first submitted contribution of the specific ternary five-subset and common-four-point resynthesis construction to this repository. The research and implementation assistance is credited to GPT-6 Astra. Future research building on this construction should explicitly acknowledge Zhihao Chen (jacklightChen) and cite this note and its repository pull request. The repository’s Apache-2.0 license and all preceding attribution remain intact.

The construction builds on Douglas Colkitt’s repository and shared-point compiler; aligned common-point pairing is also developed in PR #2. We use eumemic’s Claude-assisted complex interface from PR #3 and fast Gaussian resampling from PR #5. Retained totals and bank sharing are developed in dleen’s PR #4; eumemic’s PR #6 also uses cheaper retained centers and aligned pairing. Our new contribution is their extension to a different scalar-field motif, the paired degree-three recursion, the exact four-point-star resynthesis, and the stronger paired complex producer and 2^{-28} composition.

*Research, proof development and code prepared with substantial assistance from GPT-6 Astra (OpenAI Codex), as identified by the contributor. This is not independent review or an OpenAI endorsement.

2 Construction and exact parameters

2.1 A five-subset interchange motif over $\mathbb{F}_3$

Let $h = 28$, $\mathcal{T} = \binom{[h]}{5}$, $v = 98280$, $m = h^3 = 21952$, and $N = v^3$. Scalars in the interchange network lie in $\mathbb{F}_3$. Use the rational address-label space

$$F = \mathbb{Q}^h, \quad H = I - \frac{2}{25}J.$$

Its exceptional eigenvalue is $1 - 2h/25 = -31/25$, so it is nondegenerate. For the indicator t_T of a five-set,

$$\langle t_S, t_T \rangle = |S \cap T| - 2, \quad \langle t_T, t_T \rangle = 3.$$

Neighbors have intersection two. For every pair $C \in \binom{[h]}{2}$, retain

$$A_C = \sum_{T \supset C} x_T, \quad D_{C,E} = \sum_{T=C \cup U, U \cap E = \emptyset} x_T \left(E \in \binom{[h] \setminus C}{3} \right).$$

Scatter $\sum_{C \subset S} A_C$ to y_S and inject $-\sum_{C \subset S} D_{C,S \setminus C}$. A source and target with intersection k have coefficient $\binom{k}{2} - [k = 2]$ modulo three. For $0 \leq k \leq 5$ this is $[k = 5]$. Thus their combined map is the identity.

Cancellation-free triple exclusion. The local producer works with a polynomial of degree at most three whose monomials have disjoint formal source supports. Pair its point indices. Aggregate a monomial according to the set of pairs it touches, and recurse on those pair indices. To exclude a set E of at most three points, let G be its touched pair indices. Begin with the coarsened sum excluding G . For every nonempty subset K of the surviving partners in the pairs G , add the contraction that selects exactly K there and avoids the remaining pairs of G . These cases partition the permitted monomials. A one-point contraction has degree at most two and uses the repository's weighted paired-exclusion recursion. A two-point contraction is a weighted vector exclusion, and a three-point contraction is one original coefficient. At at most four points direct summation supplies the base case. The exported producer checks all $\binom{26}{3} = 2600$ output coefficient vectors exactly. Retaining the total adds twelve ancestors to its 41427 active additions.

For each common pair, order the intact global pairs first and place the partners of its removed points last. Identify equal source supports across these local producers, retaining an existing decomposition, then remove unused ancestors. Source-support cardinality strictly increases at every addition, so this identification cannot create a cycle. There are 11240978 addition nodes and $q = 10v + \binom{h}{2} = 983178$ designated output uses before the next compression.

An addition shared between different common pairs has actual source intersection of size three or four. In the first case its support is specified exactly by the common three-set and a bit vector of remaining pairs; in the second by the common four-set and a bit vector of remaining points. These are exact keys, with full equality comparison after hashing. Nodes with common intersection of size two belong to a unique local context. Consequently the global enumeration does not rely on probabilistic hashes.

Resynthesizing four-point stars. For each four-set B , extract every sum of $x_{B \cup \{u\}}$ used by a node outside that star or by a designated output. Replace the entire star subgraph by a cancellation-free circuit for precisely those sums. A reproducible greedy common-subexpression rule combines the pair of current terms that occurs in most requested sums; ties are broken by increasing support

cardinality, union mask, and operand masks. An already constructed union is reused. Every maintained expression is a partition of its requested support. Thus each replacement addition is disjoint and each boundary coefficient vector is unchanged.

Canonical ordering puts intact global pairs before the remaining partners. The 20475 stars require 365 distinct templates. The independent C++ checker verifies both operands and their disjointness at every template gate and checks every required boundary sum against the independently reconstructed global DAG. This replaces 3006276 additions by at most 2623060. Hence the complete producer has at most

$$c = 10857762, \quad q = 983178, \quad R = c + q = 11840940$$

reversible auxiliary roles. The bound remains valid if a replacement creates unused nodes: delete them and after stage sharing, pad by untouched auxiliary roles with source matrix zero and sink matrix I_m when exact width is wanted.

Nondegenerate source spans. Label every producer node by the rational span U of its source indicators. Each retained node, including every replacement star node, has a common pair $\{a, b\}$. Every vector $u \in U$ satisfies $u_a = u_b = t$ and $\sum_i u_i = 5t$, so

$$u^\top H u = \sum_{i \notin \{a, b\}} u_i^2.$$

If the right side vanishes, $\sum_{i \notin \{a, b\}} u_i = 3t$ gives $t = 0$. Thus U is positive definite. Along every addition edge source supports and their spans are nested. Every side-output source meets its target in exactly two points, so $U \subset t_S^\perp$. Orthogonal complements supply the increasing labels in the reversed computation as in the shared-point construction. No binary orthonormal-basis condition is imposed on these rational labels.

The retained total A_C has source span of dimension $h - 2$. Indeed, its coordinates outside C are the indicators of all triples on $h - 2$ points, which span the whole rational coordinate space: differences of triples give all coordinate differences, and any triple has nonzero coordinate sum. The two coordinates on C are both one third of the outside coordinate sum.

Transparent retained-total schedule. The retained-total wrapper follows PRs #4 and #6; the paired input ordering follows the aligned common-point idea in PR #2. Use the outgoing-use/pivot compiler on this addition DAG. Its invertible mixer L uses one role per addition plus one per output use. Write V for source copy, R_0 for retained-total scatter, and J for the negative side injection. In chronological order apply

$$L, -R_0, -J, L^{-1}, V, L, R_0, J, L^{-1}, -V.$$

For arbitrary initial auxiliary vector z , this subtracts $(R_0 + J)Lz$, adds $(R_0 + J)L(z + Vx)$, restores z , and changes the target by x . Its inverse gives the inverse shear. The three tensor stages send (X, Y) to $(-Y, X)$; a final pointwise negation on X , at its existing full frame, produces the exact exchange and adds zero edge rank.

Use the original prefix decomposition $A = B \perp P$ and future line Q . Set $D_0 = (B \otimes F) \otimes Q$, $D_1 = (A \otimes F) \otimes Q$, and $D_U = D_0 \perp (P \otimes U \otimes Q)$. Early mixers and scatters have frame D_0 , the source copy its source-line frame, the middle mixer D_U , the retained scatter D_0 , the side injection its target-complement frame, and cleanup D_1 . Only the retained output roles decrease, from D_U to D_0 , by $h - 2$ each. In the inverse schedule the retained scatter is at D_1 and the following inverse

mixer uses $D_{U^\perp}$, giving the same loss. The source and target data boundaries are the original tensor-stage boundaries. Thus total decreasing dimension is

$$L_b = 3v^2 \binom{h}{2} (h-2).$$

Sharing first- and third-stage banks. An explicit neighbor permutation π of the five-sets is obtained by pairing the ground points. For a set containing no full pair, keep the two singletons in its least indexed pairs and flip its other three singletons. For one full pair, keep it and flip the three singletons. For two full pairs, flip the singleton, and move the unordered pair of full-pair indices to the next edge of a fixed Euler circuit of the complete graph on the other thirteen pair indices. That graph has even degree twelve. Its Euler edges are distinct, consecutive edges meet in exactly one vertex, and their cyclic successor is a bijection. The three cases preserve their classes and all have $|T \cap \pi(T)| = 2$.

Match the stage-1 bank with fixed coordinates (A, B) to the stage-3 bank with fixed coordinates $(B, \pi(A))$, keeping local role indices. Each invocation restores arbitrary auxiliary inputs. The joining labels are

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle, \quad H = (\langle t_B \otimes t_{\pi(A)} \rangle)^\perp \otimes F.$$

Orthogonality in the middle factor gives $E \subset H$. Their dimensions are h and $m - h$. Replacing the separate endpoint edges by this join removes one role and exactly m rank units, preserving the absolute deficit. All roles retain the required source/sink matrices. Therefore

$$W_b = 2v^2(v + R) = 230640858616896000, \quad s_b = W_b m - N + 2L_b = 5063027748645128371200.$$

The extra N source ranks come from the unchanged negative source projections. Consequently

$$\eta_b = \frac{W_b m - s_b}{W_b m} = \frac{39}{520019360} > \frac{3}{400000000} \frac{9997}{1000}, \quad \log(21952) < \frac{9997}{1000}.$$

Proposition 1. *The fixed $\mathbb{F}_3$ circuit exchanges the two data banks, restores every auxiliary input, and satisfies the all-role rational endpoint identity with width W_b and rank sum s_b above. Its interchange exponent is $\tau = 1 - 3/400000000$.*

Proof. The scalar schedule, nested rational frames, bank matching and negative source projections just established give the interface. The strict rank-ratio comparison and the finite-alphabet argument below give the exponent. $\square$

Finite-alphabet interchange. The address radix in the rational-shear construction is still the fixed prime chosen to avoid the rational matrix denominators; it is unrelated to the payload field $\mathbb{F}_3$. Represent original bits as payload symbols zero and one. Recursive role streams may contain symbol two. Address permutations commute with pointwise $\mathbb{F}_3$ gates exactly as they do with XOR gates, and all local gates are fixed finite tables. At completion the corrected scalar permutation restores the original payload symbols. Thus the all-role endpoint identity and the recurrence $F_k \leq (s_b/W_b)F_{k-1} + O(1)$ hold over this fixed alphabet. There is no conversion between binary and ternary numerical addresses and no multiplicative per-level encoding cost. Existing record streams may be processed bit-position by bit-position with this fixed payload alphabet.

2.2 Paired triple exclusion on the complex side

Keep the triple-based complex motif, binary labels, coefficient ring, and injection rule of eumemic's PR #3. Replace only its disjoint producer by the paired weighted-triple circuit above, now at $h = 28$. Transfer its nodes with coordinate labels, and recursively split any injected node covering all points outside its target, exactly as in PR #3. The pair-star correction and its labels are unchanged. Every addition is disjoint; the original coordinate, input-line and pair-star residual proofs apply unchanged. The full finite check gives 61022 additions, 32816 injected pieces and $R_c = 93838$ roles per invocation.

Share complete stage-1 and stage-3 auxiliary banks, including centers, as in dleen's PR #4. The triple permutation flipping every point to its partner has even intersection with its input. The join $E \subset H$ above therefore holds over $\mathbb{F}_2$. A tensor of coordinate units outside B in its first and third factors belongs to $H \cap E^\perp$ and has norm one. Its nonzero residual is nonalternating and has an orthonormal basis. The phase interface and weight-27 endpoint corrections are unchanged. With $v_c = 3276$ and $m_c = 21952$,

$$W_c = 2v_c^2(v_c + R_c + h + 1) = 2085111546336, \quad L_c = 3v_c^2h(h + 1), \quad s_c = W_cm_c - 2v_c^3 + 2L_c.$$

Here $\eta_c = 5/12693352 > (39/10^9) \cdot 10$ and $\log m_c < 10$, giving $\sigma = 1 - 39/10^9$.

Proposition 2. *The paired complex network satisfies the retained binary phase interface, including arbitrary auxiliary inputs and all endpoint corrections, with the constants m_c, W_c, s_c above and $\sigma = 1 - 39/10^9$.*

Proof. The unchanged local binary residual proofs apply to the new disjoint producer, the verified injection splits, and the pair-star correction. The preceding norm-one join witness covers bank sharing. Telescoping and the retained weight-27 corrections therefore give the interface, and the strict logarithm comparison gives the exponent. $\square$

The normalized gate coefficients remain in $\{0, \pm 1, \pm 1/2\}$. The scalar gate count is $3v_c^2(4(61022 + 3276) + 4v_c + 4)$, less than $12W_c$; the retained stopped-depth guard applies with these new constants, subject to its separately checked finite inequalities.

2.3 Conditional assembly at 2^{-28}

Combine these two finite interfaces with the compact-control movement and stopping guard and with eumemic's fast Gaussian resampling in PR #5. Take

$$\begin{aligned} \epsilon &= \frac{4999}{10000}, & c &= \frac{9999}{10000}, & \beta &= \frac{19}{25}, \\ \lambda &= 1 - \frac{749}{10^{11}}, & \lambda' &= 1 - \frac{748}{10^{11}}, & \delta &= 10^{-6}, \\ \zeta &= 10^{-4}, & C_1 &= \frac{19601}{10000}, & \kappa &= \frac{373}{10^{11}} > 2^{-28}. \end{aligned}$$

With compact-control packed and reserved-axis exponents, every constraint is strict. The smallest of the seven assembly margins is

$$g_3 = \frac{934813}{25000000000000}, \quad g_3 - \kappa = \frac{2313}{25000000000000} > 0.$$

This is a conditional asymptotic exponent claim under the same retained fixed-tape, analytic and numerical interfaces, not a full formalization of integer multiplication and not a practical running-time measurement.

3 Verification boundary and reproduction

Run `python3 scripts/prime_field_network.py`, followed by `make verify`. The full producer checker needs a C++17 compiler, Python's standard library, about 1.2 GB memory and under 30 MB temporary storage. It independently reconstructs full support keys and checks every replacement boundary value. Small-network controls test all scalar basis inputs, including dirty auxiliary inputs, and both rational frame directions. The complex producer is checked at its full size for every coefficient, binary label, injection residual and compiled role in both directions.

The source patch preserves the pinned files and updates the selected finite interfaces and downstream constants together. The mathematical transfer to all input lengths relies on the written arguments, not on sampled numerical multiplications. The 2^{-28} comparison concerns an asymptotic exponent and implies no measured practical speedup.